

CHAPTER FOUR

INVENTORY MANAGEMENT

4.1. Definition of Inventory

Inventory is the physical stock of items held in any business for the purpose of future production or sales. In a production shop the inventory may be in the form of raw materials.

Although inventories are classified in many ways, the following classification is convenient for use in further discussion of the topic (Dobler and Burt, 1996):

- i. Production inventories. Raw materials, parts, and components which enter firm's product in the production process. These may consist of two general types: (1) special items manufactured to company specifications and (2) standard industrial items purchased 'off the shelf'.
- ii. MRO inventories. Maintenance, repair, and operating supplies which are consumed in the production process but which do not become part of the product (e.g., lubricating oil, soap, machine repair parts).
- iii. In-process inventories. Semi-finished products found at various steps of the production operation.
- iv. Finished goods inventories. Completed products ready for shipment.

The meaning of inventory can be defined in the following two ways (Ahuja, 1992)

- i) Inventory is defined as 'any idle resource of an enterprise'. The concept of idle resource in this definition means that it is not kept for immediate use and shows the importance of having some inventories for the smooth functioning of an organization.
- ii) Inventory is made of all those items ready for sale of items that keep the process running well. Inventory is a stock of materials that are used to facilitate production or to satisfy customers' demand. The concept of inventory control is as old as the concept of business itself. But the practical applications of inventory management got emphasis after the Second World War. The development of operations research and computer technology paved the way for the practical application of inventory management.

4.2. Significance of Inventories

There are several reasons why organizations should maintain inventories of good. The fundamental reason for doing so is that it is either physically impossible or economically unsound to have goods acquired in a given system precisely only when demands for them arise. Without inventories, customers would have to wait until their orders are to be filled from another source or until they are to be manufactured. However, customers will not or cannot allow waiting for long periods of time. For this reason alone, the carrying of inventories is necessary for almost all organizations that supply physical goods to customers. There are also other reasons for holding inventories. For instance, the price of some raw materials used by manufacturers may exhibit considerable seasonal fluctuation. When the price is low, it is profitable for organizations to procure a sufficient quantity of it to last through the high price seasons and to keep it in inventory to be used as need arises in production. Another reason for maintaining inventories especially for retail establishments is that sales and profits can be increased if one has an inventory of goods to display to customer.

Inventories serve a number of functions. Among the most important are the following (Stevenson; 2005)

1. To meet anticipated customer demand. A customer can be a person who walks in off the street to buy a new stereo system, a mechanic who requests a tool at a tool crib, or a manufacturing operation.
2. To smooth production requirements. Firms that experience seasonal patterns in demand often build up inventories during pre-season periods to meet overly high requirements during seasonal periods. These inventories are aptly named seasonal inventories. Companies that process fresh fruits and vegetables deal with seasonal inventories.

3. To decouple operations. Historically, manufacturing firms have used inventories as buffers between successive operations to maintain continuity of production that would otherwise be disrupted by events such as breakdowns of equipment and accidents that cause a portion of the operation to shutdown temporarily. The buffers permit other operations to continue temporarily while the problem is resolved. Similarly, firms have used buffers of raw materials to insulate production from disruptions in deliveries from suppliers, and finished goods inventory to buffer sales operations from manufacturing disruptions.
4. To protect against stock outs. Delayed deliveries and unexpected increases in demand increase the risk of shortages. Delays can occur because of weather conditions, supplier stock outs, deliveries of wrong materials, quality problems, and so on. The risk of shortages can be reduced by holding safety stocks, which are stocks in excess of average demand to compensate for variabilities in demand and lead time.
5. To take advantage of order cycles. To minimize purchasing and inventory costs, a firm often buys in quantities that exceed immediate requirements. This necessitates storing some or all of the purchased amount for later use. Similarly, it is economical to produce in large rather than small quantities. Again, the excess output must be stored for later use. Thus, inventory storage enables a firm to buy and produce in economic lot sizes without having to try to match purchases or production with demand requirements in the short run. This results in periodic orders, or order cycles. The resulting stock is known as cycle stock. Order cycles are not always based on economic lot sizes. In some instances, it is practical or economical to group orders and/or to order at fixed intervals.
6. To hedge against price increases. Occasionally a firm will suspect that a substantial price increase is about to occur and purchase larger-than-normal amounts to beat the increase. The ability to store extra goods also allows a firm to take advantage of price discounts for larger orders.
7. To permit operations. The fact that production operations take a certain amount of time (i.e., they are not instantaneous) means that there will generally be some work-in-process inventory. In addition, intermediate stocking of goods including raw materials, semi-finished items, and finished goods at production sites, as well as goods stored in ware-houses-leads to pipeline inventories throughout a production-distribution system.
8. To take advantage of quantity discounts. Suppliers may give discounts on large orders.

4.3. Inventory Management/Control

Inadequate control of inventories can result in both under and over stocking of items. Under-stocking results in missed deliveries, lost sales, dissatisfied customers, and production bottlenecks; overstocking unnecessarily ties up funds that might be more productive elsewhere. Inventory management has two main concerns. One is the level of customer service, that is, to have the right goods, in sufficient quantities, in the right place, at the right time. The other is the cost of ordering and carrying inventories.

The overall objective of inventory management is to achieve satisfactory levels of customer service while keeping inventory costs within reasonable bounds. Toward this end, the decision maker tries to achieve a balance in stocking. He or she must make two fundamental decisions: the timing and size of orders (i.e., when to order and how much to order).

In general, inventory control has the following major advantages (Jessop and Morrison: 1998):

- Helps in keeping the investment in the inventories as low as feasible.
- Ensures availability of materials by providing adequate protection against uncertainties of supplies and consumption of materials.
- Allows full advantage of economics of bulk purchase and transportation.
- Reduce chances of going out of stock.
- Leads to reduction in inventory levels.

- Releases more capital for other operations.
- Increase profitability of an organization.
- Helps to render adequate customer services.
- Renders advantage of price discounts from bulk purchasing.
- Provides flexibility to allow changes in production time due to changes in demands or any other reason, and
- Even-out the workloads on the shop in the face of functioning demands.

On the other hand, according to Ahuja (1992) the following issues need special consideration to get the advantage from inventory management:

- The need to minimize the existence of time lags between manufacturing and transport operations.
- The need to schedule various stages of the system independently.
- The need to meet the fluctuation in demand and production rates.
- The need to maintain control over the quality of the finished product.
- The need to exercise influence over changes of materials.

Before inventory control process starts, inventory planning should be executed. Inventory planning is the determination of the type and quantity of inventory items that would be required at future points for maintaining production schedules.

4.4. Inventory Costs

Because we are interested in optimizing the inventory system, we must determine an appropriate optimization or performance criterion. Virtually all inventory models used cost minimization as the optimization criterion. An alternative performance criterion might be profit maximization. However, cost minimizations and profit maximizations are essentially equivalent criteria for most inventory control problems. There are four major elements of inventory costs that should be taken for analysis, such as

- | | |
|--------------------|-------------------|
| (1) Item costs | (3) Holding costs |
| (2) Ordering costs | (4) Shortage cost |

1. Item (Purchased) costs

This is cost of the item whether it is manufactured or purchased. If it is manufactured, it includes such items direct material and labor, indirect materials and labor and overhead expenses. When the item is purchased, the item cost is the purchase price of one unit. Transportation costs, which also affect the lot size, often are included in the purchase cost of the materials.

2. Ordering (setup) costs

These are fixed costs usually associated with the production of a lot internally or the placing of an order externally with a vendor. In other words, these costs are independent of the number of units that are requested. Setup costs are related to the amount of time needed to adjust the equipment to perform a specific task. This would include the alignment of special tooling such as jigs and fixtures. Order costs pertain to the costs involved in placing an order with a vendor. These may include telephone charges, a delivery fee, expediting costs, and the time required to process a purchase order.

3. Holding (or carrying) costs

If the item is held in stock, the cost involved is the item carrying or holding cost. Carrying material in inventory is expensive. A number of studies disclosed that the annual cost of carrying a production inventory average approximately 25 percent of the value of the inventory. Let us briefly examine these carrying costs (Dobler and Burt; 1996).

- (i) Opportunity cost of invested funds. When a firm purchases \$50,000 worth of a production material and keeps it in inventory, it simply has this much less cash to spend for other purposes. Money

invested in productive equipment or in external securities earns a return for the company. Conceptually, then, it is logical for the firm to charge all money invested in inventory an amount equal to that it could earn if invested elsewhere in the company. This is the “opportunity cost” associated with inventory investment.

- (ii) Insurance costs. Most firms insure their assets against possible loss from fire and other forms of damage. An extra \$50,000 worth of inventory represents an additional asset on which insurance premiums must be paid.
- (iii) Property taxes. As with insurance, property taxes are levied on the assessed value of a firm’s assets; the greater the inventory value, the greater the asset value, and consequently the higher the firm’s tax bill.
- (iv) Storage costs. The warehouse in which a firm stores its inventory is depreciated a certain number of dollars per year over the length of its life. One may say, then, that the cost of warehouse space is a given number of dollars per cubic foot per year. And this cost conceptually can be charged against inventory occupying the space.
- (v) Obsolescence and deterioration. In most inventory operations, a certain percentage of the stock spoils, is damaged, is pilfered, or eventually becomes obsolete. No matter how diligently warehouse managers guard against these occurrences, a certain number always takes place. With new products being introduced at an increasing rate, the probability of obsolescence is increased accordingly. Consequently, the larger the inventory, typically the greater the absolute loss from this source.

Other types of costs included in the unit holding cost are:

- Cost of maintaining inventory records (stationary and other consumables used by the stores)
- The salaries and wages of storing, receiving and issue of material personnel.

NB: 1. Holding costs are stated in either of two ways: as a percentage of unit price or as a birr amount per unit. However, as we generally measure inventory in unit rather than in birr, it is convenient to express the holding cost in terms of birr per unit per year rather than per birr per year.

2. If holding cost per item is given for a period of less than a year, you need to convert the same into cost per year.

(4) Shortage (penalty or Stock-out) costs.

Shortage costs result when demand exceeds the supply of inventory on hand. These costs can include the opportunity cost of not making a sale, loss of customer goodwill, late charges, and similar costs. Furthermore, if the shortage occurs in an item carried for internal use (e.g., to supply an assembly line), the cost of lost production or downtime is considered a shortage cost. Such costs can easily run into hundreds of birrs a minute or more. Shortage costs are sometimes difficult to measure, and they may be subjectively estimated.

4.5. Dependent and Independent Demand

Independent demand: - the demands for various items are unrelated to each other and therefore the required quantities of each must be determined separately or independently.

Dependent demand: - the requirement for any one item is a direct result of the need for some other item, usually a higher-level item of which it is a component or subassembly.

In concept, dependent demand is a relatively straight forward computational problem. The required quantities of a dependent-demand item are simply computed, based on the number needed in each higher-level item where it is used. For example, if an automobile company plans on producing 500 automobiles per day, then obviously it will need 2,000 wheels and tires (plus spares). The number of wheels and tires needed is dependent on the production level for automobiles and not derived separately. The demand for automobiles, on

the other hand, is independent-it comes from many sources external to the automobile firm and is not a part of other products and so is unrelated to the demand for other products.

4.6. Inventory Systems for Independent Demand

If one has to make decisions about managing an inventory, it is useful to understand the behavior of the inventory related cost factors discussed in the previous sections. These factors often help a manager to determine which items should or should not be carried in inventory, what inventory levels should be carried for specific items, and what order quantities are appropriate for given items. There are a number of mathematical models that can be applied to determine the optimum (economical) level for independent demand materials. Some of these models are discussed in this section.

4.6.1. Economic Order Quantity (EOQ) Model

The EOQ (Economic Order Quantity) model is one method of determining the adequate (optimum) inventory level for independent demand materials. It is used to identify a fixed order size that will minimize the sum of the annual costs of holding inventory and ordering inventory. This model involves a number of assumptions including the following.

1. Only one product is involved.
2. Annual demand requirement are known.
3. Demand is spread evenly throughout the year so that the demand rate is reasonably constant.
4. Lead time does not vary.
5. Each order is received in a single delivery.
6. There are no quantity discounts.

Figure 4.1 below shows clearly that as the order or delivery quantity increases, carrying costs rise-and at the same time ordering costs decrease. Carrying cost is thus a linear function of Q : carrying costs increase or decrease in direct proportion to changes in the order quantity Q , as the figure illustrates. On the other hand, annual ordering cost will decrease as order size increases because, for a given annual demand, the larger the order size, the fewer the number of orders needed. Unlike carrying costs, ordering costs are relatively insensitive to order size; regardless of the amount of an order, certain activities must be done, such as determining how much is needed, periodically evaluating sources of supply, and preparing the invoice. Even inspection of the shipment to verify quality and quantity characteristics is not strongly influenced by order size since large shipments are sampled rather than completely inspected. Hence, ordering cost is treated as a constant.

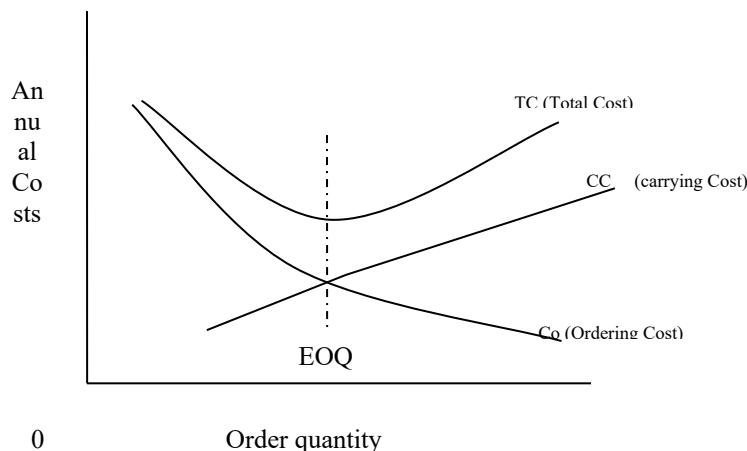


Fig. 4.1 Relationship of inventory-related costs to inventory level

In constructing any inventory model, the first step is to develop a functional relationship between the variables of interest and the measure of effectiveness. In this case, since we are concerned with cost, the following equation would pertain:

$$\text{Total Annual cost} = \text{Annual Ordering cost} + \text{Annual Holding cost} + \text{Annual Purchase cost}$$

To develop an equation for total inventory cost and for the purpose of analyzing inventory models, the following symbols will be used throughout the chapter.

TC = Total annual cost

D = Annual demand in units

Cc = Carrying cost per unit

C_o = Set up or Ordering cost

Q = Quantity to be ordered

P = Purchase price per unit or cost per unit.

NB: D and Cc must be in the same units, e.g., months, years.

$$\text{Annual Ordering cost} = \left(\frac{\text{Number of orders Placed per year}}{\text{Ordering cost per order}} \right) \times$$

The number of orders per year will be D/Q, and hence:

$$\text{Annual Ordering cost} = \frac{D}{Q} C_o$$

$$\begin{aligned} \text{Annual Holding cost} &= (\text{average inventory value}) \times (\text{inventory carrying cost as a \% of inventory value}) \\ &= (\text{average inventory in units}) \times (\text{material unit cost}) \times (\text{inventory carrying cost as a \% of inventory value}) \end{aligned}$$

The average inventory is simply half of the order quantity, Q/2, and hence:

$$\text{Annual holding cost} = \frac{Q}{2} C_c$$

The annual purchase cost for the units is the product of annual demand in units and cost per unit, hence:

$$\text{Annual purchase cost} = DP$$

Bringing those costs together, we have the following total cost equation.

$$TC = \frac{D}{Q} C_o + \frac{Q}{2} C_c + DP$$

The next step is to find that order quantity, Q , for which total cost is a minimum. This can be done in two ways:

i. **Using calculus.** As depicted in figure 4.1, the total cost is minimum at the point where the slope of the total cost curve is zero. Hence, the procedure is taking the first derivative of total cost with respect to Q (which is the slope) and setting this equal to zero.

$$TC = \frac{D}{Q} C_o + \frac{Q}{2} C_c + DP$$

$$TC = D \cdot Q^{-1} \cdot C_o + \frac{Q}{2} C_c + DP$$

$$\text{Slope (MC)} = \frac{dTC}{dQ} = -D \cdot Q^{-2} \cdot C_o + \frac{1}{2} C_c + 0$$

Setting this equal to zero, we have:

$$-D \cdot Q^{-2} \cdot C_o + \frac{1}{2} C_c = 0$$

$$-\frac{D}{Q^2} \cdot C_o + \frac{C_c}{2} = 0$$

$$-\frac{D}{Q^2} \cdot C_o = -\frac{C_c}{2}$$

$$\frac{D \cdot C_o}{Q^2} = \frac{C_c}{2}$$

$$Q^2 \cdot C_c = 2DC_o$$

$$Q^2 = \frac{2DC_o}{C_c}$$

$$Q = \sqrt{\frac{2D \cdot C_o}{C_c}}$$

ii. **Equating ordering and carrying costs.** Figure 4.1 also shows that the total cost is minimum at a point where holding cost crosses ordering cost (holding cost equal to ordering cost). Therefore, by equating these two costs the formula to get the economic order quantity can be arrived at as shown below.

$$\left(\frac{D}{Q} C_o\right) = \left(\frac{Q}{2} C_c\right)$$

$$D \cdot C_o = \frac{Q^2}{2} \cdot C_c \quad (\text{Multiplying both sides by } Q)$$

$$2D \cdot C_o = Q^2 \cdot C_c \quad (\text{Multiplying both sides by } 2)$$

$$Q^2 = \frac{2D \cdot Co}{Cc}$$

$$Q = \sqrt{\frac{2 \cdot D \cdot Co}{Cc}}$$

This formula is the fundamental mathematical representation of the EOQ concept. The EOQ concept can be used in conjunction with a variety of inventory management systems including Just in Time (JIT). The costs which are relevant in making an EOQ analysis are incremental costs.

Incremental costs are those costs that actually change as a result of a particular operating decision. For example, if the decision is to issue more purchase orders during the year actually increases supply and service costs also increases. These are incremental costs. Incremental costs are either reliable costs or opportunity costs that represent a forgone opportunity to utilize an asset in some other productive way.

Even though the model has been set in the purchasing environment, the EOQ concept has broader application as well. It can also be used to determine the economic production lot sized in a manufacturing operation. In converting the formula for production use, the annual usage or demand and carrying cost factors are the same as they were in the purchasing application. The unit cost factor, however, is no longer delivered price. It consists of direct labor and materials and production overhead costs. Production acquisition cost is similar to purchasing ordering cost.

Example 1:

A local distributor for Addis Tire expects to sell approximately 9,600 steel-belted radial tires of a certain size and tread design next year. Annual carrying cost is Birr 16 per tire, and ordering cost is birr 75. The distributor operates 288 days a year. The purchase price of the tires is birr 900 each.

- a. What is the EOQ?
- b. How many tires per year does the store reorder?
- c. What is the length of an order cycle?
- d. What is the total annual cost if the EOQ quantity is ordered?

Example 2:

Tebarek and Family Untied assemble Vestel Television. It purchases 3,600 color cathode ray tubes a year at birr 65 each. Ordering costs are birr 31, and annual carrying costs are 20 percent of the purchase price. Compute the optimal quantity and the total annual cost of ordering, carrying and purchasing the inventory.

4.6.2. EOQ with Quantity Discount and Price Breaks

The concept of Economic Order Quantity fails in certain cases where there is a discount offered when purchases are made in large quantities. Certain manufacturers offer reduced rate for items when a larger quantity is ordered. It may appear that the inventory **holding cost may increase** if large quantities of items are ordered. **But if the discount offered** is so attractive that it even outweighs the holding cost, then probably the order at levels other than the EOQ would be economical. In such cases, the optimum quantity is the one that makes the savings from the purchase cost equal to the total inventory cost.

The procedures to be followed to calculate the optimal order quantity in a situation that involves quantity discount and price breaks are as follows:

- i. For carrying costs that are constant, the procedure for determining the optimal order quantity is as follows:

1. Compute the common minimum point, EOQ point.
2. Identify the range in which the EOQ falls.
3. If the feasible minimum point is on the lowest price range, that is the optimal order quantity.
4. If the feasible minimum point is in any other range, compute the total cost for the minimum point and for the price break of all lower unit costs.
5. Compare the total costs.
6. The quantity that yields the lowest total cost is the optimal order quantity.
 - ii. For carrying costs that are expressed as percentage of price, the procedure is:
 1. Begin with the lowest unit price; compute the minimum point (EOQ) for each price range until you find a feasible minimum point (EOQ).
 2. If the minimum point for the lowest unit price is feasible, it is the optimal order quantity.
 3. If the minimum point is not feasible in the lowest price range, compare the total cost at the price break for all lower prices with the cost of EOQ.
 4. The quantity which yields the lowest total cost is the optimum.

Example 1:

A producer of photo equipment buys lenses from a supplier at Birr 100 each. The producer requires 125 lenses per year, and the ordering cost is Birr 18 per order. Carrying costs per unit per year are estimated to be Birr 20 each. The supplier offers 6% discount for purchases of 50 lenses and an 8% discount for purchases of 100 or more lenses. Determine the optimal order quantity.

Advantages and Limitations of EOQ

Limitations of EOQ are imposed by its underlying assumptions as discussed above. The EOQ can be determined in the above manner only if the assumed conditions are in operation. But in real-life situations, none of these conditions are found to be operating. Lead times vary; rate of usage, unit price, and carrying costs are never constant. Moreover, companies often find it profitable to take advantage of quantity discounts. Finally, safety stocks are needed to prevent the occurrence of stock out contingencies due to delayed deliveries as well as higher than assumed rate of usage. The use of EOQ is also limited by the difficulty of estimating ordering and carrying costs. Nevertheless, the model serves a useful purpose on an approximation of real-life situations. Understanding of the mechanics of the model and its limitations also helps in determining economic lot size of manufactured as well as purchased items in real-life situations.

The advantages of economic ordering can broadly be described as:

1. Substantial reduction in the number of purchase orders issued.
2. Materials department have more time to devote to the items, which contain large amount of money and concentrate on the inventory control of these items.
3. The use of this formula helps to reduce possibility of stock out which may otherwise would result in more sever losses and eliminate rush purchases with attendant problems.
4. It will require lesser amount of follow-up action because of better planning or purchases, as the system is efficient, only a few hands are required.

4.6.3. Reorder Point (ROP)

EOQ model answer the question of how much to order, but not the question of when to order. The latter is the function of models that identify the reorder point (ROP) in terms of a quantity. The reorder point occurs when the quantity on hand drops to a predetermined amount. That amount generally includes expected demand during lead time and perhaps an extra cushion of stock, which serves to reduce the probability of experiencing a stock out during lead time. Lead-time is defined as the time interval between the placing of the orders and the

actual receipt of goods. Note that in order to know when the reorder point has been reached, a perpetual inventory is required.

The goal in ordering is to place an order when the amount of inventory on hand is sufficient to satisfy demand during the time it takes to receive that order (i.e., lead time). There are four determinants of the reorder point quantity:

1. The rate of demand (usually based on a forecast).
 2. The lead time.
 3. The extent of demand and/or lead time variability.
 4. The degree of stock out risk acceptable to management
- i. *ROP when demand and lead time are both constant.* If demand and lead time are both constant, the reorder point is simply

$$\text{ROP} = d \times \text{LT}$$

Where: d = Demand rate (units per day or week)

LT = Lead time in days or weeks.

Note: Demand and lead time must be expressed in the same time units.

Example1:

Mr. X takes Two-a-Day vitamins, which are delivered to his home by a route man seven days after an order is called in. At what point should Mr. X reorder?

- ii. Reorder point when variability is present in demand or lead time. Variability in demand or lead time creates the possibility that actual demand will exceed expected demand. Consequently, it becomes necessary to carry additional inventory, called safety stock, to reduce the risk of running out of inventory (a stock out) during lead time. The reorder point then increases by the amount of the safety stock:

$$\text{ROP} = \frac{\text{Expected demand}}{\text{during lead time}} + \text{Safety stock}$$

Because it costs money to hold safety stock, a manager must carefully weigh the cost of carrying safety stock against the reduction in stock out risk it provides. The customers service level increase as the risk of stock out decrease. Order cycle service level can be defined as the probability that demand will not exceed supply during lead time. Hence, a service level of 95 percent implies a probability of 95 percent that demand will not exceed supply during lead time. The risk of a stock out is the complement of service level; a customer service level of 95 percent implies a stock out risk of 5 percent. That is,

$$\text{Service level} = 100 \text{ percent} - \text{stock out risk.}$$

The amount of safety stock that is appropriate for a given situation depends on the following factors.

1. The average demand rate and average lead time.
2. Demand and lead time variability.
3. The desired service level.

For a given order cycle service level, the greater the variability in either demand rate or lead time, the greater the amount of safety stock that will be needed to achieve that service level. Selection of a service level may reflect stock out costs (e.g., lost sales, customer dissatisfaction) or it might simply be a policy variable (e.g., the manager wants to achieve a specified service level for a certain item).

Several models can be used in cases when variability is present. The models generally assume that any variability in demand rate or lead time can be adequately described by a normal distribution. The value of z

used in a particular instance depends on the stock out risk that the manager is willing to accept. Generally, the smaller the risk the manager is willing to accept, the greater the value of z .

The first model can be used if an estimate of expected demand during lead time and its standard deviation are available. The formula is:

$$= \frac{\text{Expected demand}}{\text{during lead time}} + z\delta_{dLT}$$

Where: z = Number of standard deviations.

δ_{dLT} = The standard deviation of lead time demand.

Example 2:

Suppose that the manager of a construction supply house determined from historical records that demand for sand during lead time averages 50 tons. In addition, suppose the manager determined that demand during lead time could be described by a normal distribution that has a mean of 50 tons and a standard deviation of 5 tons. Answer these questions, assuming that the manager is willing to accept a stock out risk of no more than 3 percent:

- a. What value of z is appropriate?
- b. How much safety stock should be held?
- c. What reorder point should be used?

4.6.4. Economic Production Quantity (Economic Run Lengths)

We have seen the application of the EOQ model in determining the optimum order quantity of items purchased/ordered from external suppliers. But when the company is the producer and user of its items, the run size is the economic production quantity (EPQ). When a firm is producing its own inventory rather than purchasing it, the order cost is replaced by manifesting setup costs. Setup costs are of a fixed nature like order costs. They represent the onetime costs for machine adjustments, paperwork, scheduling efforts, and the like.

The assumptions of the EPQ model are similar to those of the EOQ model, except that instead of orders received in a single delivery, units are received incrementally during production. The assumptions are:

1. Only one item is involved.
2. Annual demand is known.
3. The usage rate is constant.
4. Usage occurs continually, but production occurs periodically.
5. The production rate is constant.
6. Lead time does not vary.
7. There are no quantity discounts.

The EOQ equation must also be modified to account for the time it takes to produce item of which only a portion go in to inventory. The other portion is used concurrently in the production process or is sold as produced. The production rate, P , is of course non instantaneous and must be greater than or equal to the demand rate.

The modification to the EOQ equation for non-instantaneous supply does not affect the ordering cost (which now becomes the set up cost). But it acts to reduce carrying cost by deleting the carrying charge on that portion of the production that does not go into inventory.

If d = demand per day and P = production per day, then the ratio d/p represents the proportion of production that is allocated to daily demand, and $1-d/p$ represents that portion of the production run that goes in to inventory. If we take into account the decreased carrying cost of this reduced level of inventory, the economic

run length (ERL) or the economic production quantity (EPQ) number of units to produce per production set up is

$$EPQ = \sqrt{\frac{2 \cdot D \cdot S}{Cc(1 - d/p)}}$$

Where:

S = Set up cost in birr/set up

D = annual demand in units

Cc = Carrying costs in birr/unit/year

d = demand rate

P = production rate

Example 1:

A toy manufacturer uses 48,000 rubber wheels per year for its popular dump truck series. The firm makes its own wheels, which it can produce at a rate of 800 per day. The toy trucks are assembled uniformly over the entire year. Carrying cost is Birr 1 per wheel a year. Setup cost for a production run of wheels is birr 45. The firm operates 240 days per year.

Determine the:

- Optimal run size.
- Minimum total annual cost for carrying and setup.
- Cycle time for the optimal run size.
- Run time.

4.6.5. Single-Period Model

The single-period model (sometimes referred to as the news boy problem) is used to handle ordering of perishables (fresh fruits, vegetables, seafood, cut flowers) and items that have a limited useful life (newspapers, magazines, spare parts for specialized equipment). The period for spare parts is the life equipment, assuming that the parts cannot be used for other equipment. What sets unsold or unused goods apart is that they are typically carried over from one period to the next, at least not without penalty. Day-old baked goods, for instance, are often sold at reduced prices; leftover seafood may be discarded; and out-of-date magazines may be offered to used book stores at bargain rates. There may even be some cost associated with disposal of leftover goods.

Analysis of single-period situations generally focuses on two costs: shortage and excess. Shortage cost may include a charge for loss of customer goodwill as well as the opportunity cost of lost sales. Generally, shortage cost is simply unrealized profit per unit. That is

$$C_{\text{shortage}} = C_s = \text{Revenue per unit} - \text{Cost per unit.}$$

If a shortage or stock out relates to an item used in production or to a spare part for a machine, then shortage cost refers to the actual cost of lost production.

Excess cost pertains to items left over at the end of the period. In effect, excess cost is the difference between purchase cost and salvage value. That is

$$C_{\text{excess}} = C_e = \text{Original cost per unit} - \text{Salvage value per unit.}$$

If there is cost associated with disposing of excess items, the salvage will be negative and will therefore increase the excess cost per unit.

The goal of the single-period is to identify the order quantity, or stocking level, that will minimize the long-run excess and shortage costs. There are two categories of problems that we will consider: those for which demand can be approximated using a continuous distribution (such as uniform or normal distribution) and those for which demand can be approximated using a discrete distribution (such as historical frequencies or the Poisson).

The kind of inventory can indicate which type of model might be appropriate. For example, demand for petroleum, liquids, and gases tend to vary over some continuous scale, thus lending itself to description by a continuous distribution. Demand for tractors, cars, and computers is expressed in terms of the number of units demanded and lends itself to description by a discrete distribution.

Continuous Stocking Levels

The concept of identifying an optimal stocking level is perhaps easiest to visualize when demand is uniform. Choosing the stocking level is similar to balancing a seesaw, but instead of a person on each end of the seesaw, we have excess cost per unit (C_e) on one end of the distribution and shortage cost per unit (C_s) on the other. The optimal stocking level is analogous to the fulcrum of the seesaw; the stocking level equalizes the cost weights, as illustrated in Figure 4.2.

The service level is the probability that demand will not exceed the stocking level, and computation of the service level is the key to determining the optimal stocking level, So .

$$\text{Service level} = \frac{C_s}{C_s + C_e}$$

Where: C_s = Shortage cost per unit

C_e = Excess cost per unit

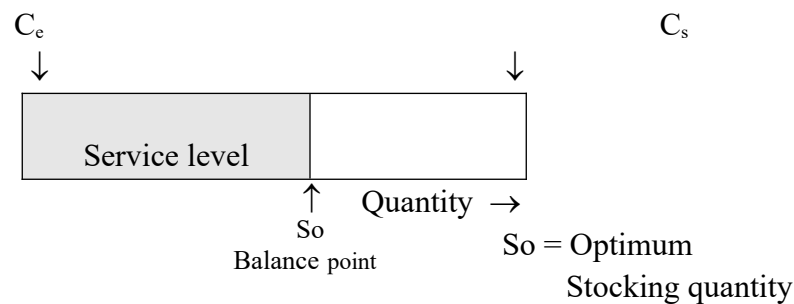


Fig. 4.1 The optimal stocking level balances unit shortage and excess costs.

If actual demand exceeds So , there is a shortage; hence, C_s is on the right end of the distribution. Similarly, if demand is less than So , there is an excess, so C_e is on the left end of the distribution. When $C_e = C_s$, the optimal stocking level is halfway between the endpoints of the distribution. If one cost is greater than the other, So will be closer to the larger cost.

Example 1:

Mekonnen Bar sells, among others, a blend of cherry juice and apple cider. Demand for the blend is approximately normal, with a mean of 200 liters per week and a standard deviation of 10 liters per week. $C_s = 60$ cents per liter, and $C_e = 20$ cents per liter. Find the optimal stocking level for the apple-cherry blend.

Fixed-Order-Interval Model

The fixed-order-interval (FOI) model is used when orders must be placed at fixed time intervals (weekly, twice a month, etc). The timing of orders is set. If demand is variable, the order size will tend to vary from cycle to cycle. This is quite different from an EOQ/ROP approach in which the order size generally remains fixed from cycle to cycle, while the length of the cycle varies (shorter if demand is above average and longer if demand is below average).

Like the ROP model, the fixed-interval model can have variations in demand only in, lead time only, or in both demand and lead time. However, for the sake of simplicity and because it is perhaps the most frequently encountered situation, the discussion here will focus only on variable demand and constant lead time.

Order size in the fixed-interval model is determined by the following computation:

$$\begin{aligned} \text{Amount to order} &= \begin{array}{c} \text{Expected demand} \\ \text{during protection} \\ \text{interval} \end{array} + \begin{array}{c} \text{Safety} \\ \text{stock} \end{array} - \begin{array}{c} \text{Amount on} \\ \text{hand at reorder} \\ \text{time} \end{array} \\ &= \bar{d}(OI+LT) + z\delta d\sqrt{OI+LT} - A \end{aligned}$$

Where:

OI = Order interval (length of time between orders)

A = Amount on hand at reorder time.

As in previous models, we assume that demand during the protection interval is normally distributed.

Example1:

Given the following information, determine the amount to order.

$\bar{d}$ = 30 units per day Desired service level = 99 percent
 δp = 3 units per day Amount on hand at reorder time = 71 units
 LT = 2 days OI = 7 days
 Z = 2.33 for 99 percent service level

$$\begin{aligned} \text{Amount to order} &= \bar{d}(OI+LT) + z\delta d\sqrt{OI+LT} - A \\ &= 30(7+2) + 2.33(3)\sqrt{7+2} - 71 \\ &= \underline{220 \text{ units}} \end{aligned}$$